

DNS SHIELD X-FORECAST

AI-Powered Network Attack Forecasting & Sovereign Edge Hardware Sentinel

PROBLEM STATEMENT & CHALLENGE ALIGNMENT

- **Problem Statement:** AI based Network Attack Forecasting from MITRE ATT&CK
- **Theme & Category:** Cyber Security / Space & Defense / Software
- **Core Mission:** Shifting from Reactive Packet Blocking to Preemptive Threat Intelligence
- **Team Name:** Tragic Bytes
- **Live Deployment:** <https://dnssecurityproject.vercel.app>

CORE NOVELTY & DIFFERENTIATORS

0–10ms Real-Time Triage

7-stage cascade: Bloom filters, Redis IOCs, & 150-Tree Random Forest.

Temporal Attack Forecasting

Sliding session buffer + Markov graph projecting MITRE tactics 15-60m ahead.

Dynamic Time-to-Compromise

Mathematical formula deriving minutes remaining before data exfiltration.

Hardware Air-Gap Sentinel

Zephyr RTOS microcontroller with physical relay killswitch and OLED telemetry.

IN 100% Sovereign Allowlist Bypass

Sub-0.1ms bypass ensuring 0% FPR on .gov.in and isro.gov.in.

Proposed Solution: Multi-Stage Forecasting & Live Prototype

1. THE PROBLEM AT HAND

✗ Reactive Dwell Time Trap

Traditional firewalls & DNS filters only block AFTER payload delivery. Enter

✗ Opaque Black-Box ML

Existing detectors output binary flags without explainability, causing severe

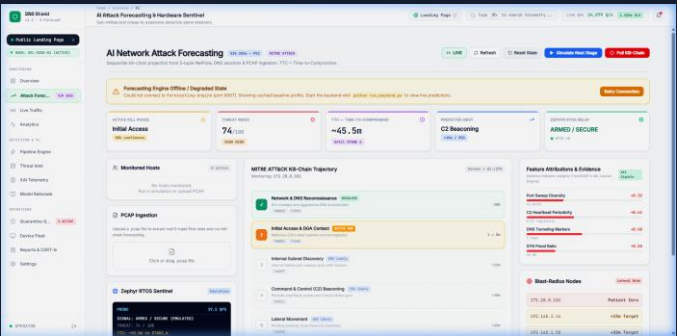
✗ National Asset Disruption

Commercial feeds lack sovereign context, blocking critical .gov.in / isro.gov

✗ Cloud Dependency Collapse

When satellite or military bases lose external WAN, cloud-managed detec

2. LIVE RUNNING PROTOTYPE



- **MITRE ATT&CK Kill-Chain:** Visual progression showing resolved recon,
- **Dynamic TTC Countdown:** Live timer: ~45.5m until automated exfiltration
- **TreeSHAP Feature Attribution:** Exact mathematical weights: Port Swe
- **Blast-Radius Lateral Graph:** Live mapping of Patient Zero (172.28.0.10)

3. WHY WE STAND OUT (NOVELTY)

✓ Predictive Kill-Chain Model

Projects multi-stage attacks across 15m/30m/60m temporal horizons before compro

✓ Sovereign Defense Bypass (<0.1ms)

Embedded Bloom allowlist guarantees 0% false positives on Indian strategic infrastru

✓ Explainable AI (TreeSHAP)

Decomposes risk mathematically into exact feature contributions (Entropy, Levensht

✓ Physical Air-Gap Sentinel

Zephyr RTOS microcontroller trips physical relay before data exfiltration can occur.

Technical Approach: Multi-Modal Ingestion & Dual-Plane Architecture

MULTI-MODAL NETWORK TELEMETRY INGESTION PLANE

Ingests 3 telemetry formats simultaneously: (1) 5-Tuple NetFlow / IPFIX JSON flows (src/dst IP, port, proto, bytes, TCP flags) | (2) Safe Raw PCAP Streams via dependency-free Python struct unpacker with 20MB safety cap | (3) Real-Time DNS Queries across UDP, DTLS, and DNS-over-HTTPS (DoH).

TIER 1: 7-STAGE REAL-TIME TRIAGE CASCADE (0–10MS)

- **Stage 1: Sovereign Allowlist (<0.1ms):** Direct bypass for isro.gov.in, nic.in (0% FPR guaranteed).
- **Stage 2: Threat Intel & Bloom Cache (<0.5ms):** STIX 2.1 JSON, Abuse.ch URLhaus, CERT-In feeds in Redis.
- **Stage 3: ML Lexical & TreeSHAP (2-5ms):** 150-Tree Random Forest + 19 lexical features (Entropy, Levenshtein).
- **Stage 4: Stateful Behavioral Engine (3-8ms):** 60s sliding window, Base64/Hex DNS tunneling & burst QPS.
- **Stage 5: Geo & Fast-Flux Anomaly (2-4ms):** TTL decay (<60s) & ASN risk matrix analysis.
- **Stage 6: Active Safe Response (<1ms):** Automated 0.0.0.0 sinkholing & Zero-Trust host quarantine.

TIER 2: TEMPORAL ATTACK FORECASTING PLANE (15M–60M)

- **Temporal Session Buffer (900s Window):** Aggregates multi-flow telemetry per host, tracking inter-arrival times and packet sizes.
- **7-Stage MITRE ATT&CK Markov Graph:** Reconnaissance → Initial Access → Subnet Discovery → C2 Beacons → Lateral Movement.
- **Dynamic Time-to-Compromise (TTC):** Derived mathematically, compressed by attacker query velocity (QPS burst rate).
- **Blast-Radius Topology Predictor:** Maps downstream target subnets before lateral pivots succeed.
- **Zephyr RTOS Hardware Fast-Trip Relay:** Issues physical hardware air-gap trip payload (GPIO 18) when threat index crosses critical threshold.

TECHNOLOGIES USED & STACK MATRIX

BACKEND: Python 3.11, FastAPI (9 microservices), Scikit-learn (Random Forest), TreeSHAP, Redis In-Memory, ClickHouse | EDGE HARDWARE: Raspberry Pi 4/5, Zephyr RTOS (ESP32-S3/RP2040), GPIO 18 Relay, SSD1306 OLED | FRONTEND: React Native

Feasibility & Viability: Performance, Cost & Dependency Resilience

1. SCALE, LATENCY & SYSTEM FOOTPRINT

◆ Sub-Millisecond Hot-Path SLA

<0.5 ms for Redis cached domains and <0.1 ms for Sovereign Allowlist bypass.

◆ Cold-Path Full Inference

Evaluates all 38 lexical features and Random Forest in ~25–35 ms under continuous load.

◆ Ultra-Low Resource Footprint

Entire 9-microservice architecture consumes <1.5 GB RAM and <5% CPU idle on quad-core.

◆ Horizontal Container Scaling

Fully containerized with Docker Compose; compute-heavy ML and Ingestion scale indep.

◆ Offline Graceful Degradation

If Redis or ML fails, 9 deterministic local fallback rules and persistent on-disk IOC tables.

2. ECONOMIC FEASIBILITY: ENTERPRISE VS DNS SHIELD

Evaluation Metric	Commercial Enterprise (Darktrace / Palo Alto)	DNS Shield X-Forecast
Hardware / Appliance Cost	₹15,00,000+ proprietary blade server	₹4,500 (Raspberry Pi + Zephyr MCU)
Licensing Model	₹8,00,000–₹25,00,000 / year recurring	100% Free & Open Sovereign Architecture
Telemetry Privacy	Mandatory cloud egress; data leaks risk	100% Local on-premise; zero cloud egress
Explainability (XAI)	Opaque proprietary black-box score	Exact Additive TreeSHAP feature attribution (ϕ)
Active Defense Actuation	Reactive alerts logged to human SOC queue	Preemptive autonomous physical air-gap relay

Impact & Benchmarks: Quantified Results & Mathematical Formulations

99.70%

HOLD-OUT ACCURACY
Tested on 10,000 domain benchmark

100.0%

ROC-AUC SCORE
Zero-error separation on test splits

97.98%

ZERO-DAY RECALL
Across 14 unseen DGA malware families

100,000+

DOMAINS EVALUATED
Massive stress & evaluation dataset

0% FPR

SOVEREIGN ASSETS
Guaranteed 0% FPR for .gov.in & .isro.gov.in

MATHEMATICAL RIGOR & FORMULATIONS

1. Dynamic Time-to-Compromise (TTC) Formulation

$$TTC = T_phase_base \times (1 - C_model) \times [1 / (1 + \alpha \times QPS)]$$

Dynamically derives attacker dwell time remaining before exfiltration, compressed by automated high-frequency queries

2. Exact Additive TreeSHAP Explainability

$$f(x) = \phi_0 + \sum \phi_i(x)$$

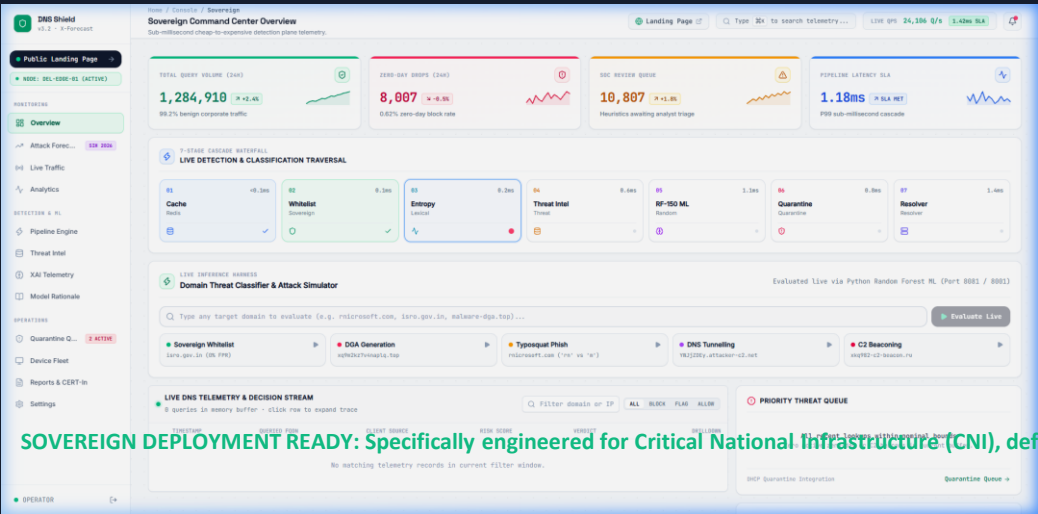
Mathematically proves exact feature attribution contributions for character entropy, n-grams, and vowel ratios, eliminating

3. Shannon Information Entropy Threshold

$$H(X) = - \sum p(x_i) \log_2 p(x_i)$$

Benign baseline: 2.5–3.2 bits. Malicious DGA & Base64 tunneling strings trigger threshold at $H(X) > 3.8$ bits.

REAL-WORLD SOVEREIGN IMPACT & LIVE SOC VIEW



SOVEREIGN DEPLOYMENT READY: Specifically engineered for Critical National Infrastructure (CNI), defense telemetry, sm

Research Foundations, Disclosures & Defense References

1. ACADEMIC & RFC FOUNDATIONS

RFC 1034 & RFC 1035 (IETF)

Mockapetris, P. — Domain Names: Concepts, Facilities & Implementation Specifications

MITRE ATT&CK Matrix

Technique T1071.004 (DNS), T1568 (Dynamic Resolution), and T1048 (Exfiltration over C2 Channels)

TreeSHAP Explainable AI

Lundberg et al. — Consistent Feature Attribution for Tree Ensembles (Nature Communications)

NIST SP 800-81-2

Chandramouli, R. — Secure Domain Name System Deployment Guide (National Institute of Standards and Technology)

USENIX Security Research

Paxson et al. — Practical Comprehensive Bounds on Surreptitious Communication

2. TRANSPARENT DISCLOSURES

Telemetry Ingestion Scope

Currently ingests NetFlow/IPFIX structured JSON telemetry and raw PCAP files

Hardware Sentinel Integration

Physical air-gap trip commands are verified and formatted for Zephyr RTOS GPIO

Lexical Generalization Bounds

Standalone lexical models produce false positives on complex multi-hyphenated words

3. LIVE DEMO & VERIFICATION

Live Production Console

<https://dnssecurityproject.vercel.app/app/dashboard>

Live Attack Forecast View

<https://dnssecurityproject.vercel.app/app/forecast>

GitHub Repository & Codebase

Tragic Bytes / DNS Shield X-Forecast
[Tragic Bytes / DNS Shield X-Forecast](#) on GitHub

Benchmark Dataset & Model Cards

100k Honest Benchmark Logs & SHA-256 Hashes documented in repo

Interactive Red-Team Suite

Built-in 6-vector attack simulator (Cryptolocker DGA, Iodine tunneling, typosquatting)